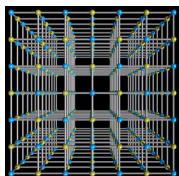


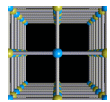
Numerical Analysis of Fracture Evolution Using Nonlocal Models



Prashant Kumar Jha
prashant.j16o@gmail.com

Joint work with
Dr. Robert Lipton

Funded by
Army Research Office



Peridynamics model with bond-based potential

1

Peridynamics equation: $\rho \ddot{\mathbf{u}}(\mathbf{x}, t) = -\nabla PD^\epsilon(\mathbf{u}(t))(\mathbf{x}) + \mathbf{b}(\mathbf{x}, t)$

Strain: Linearized strain (assuming small displacement)

$$S(\mathbf{y}, \mathbf{x}; \mathbf{u}) = \frac{\mathbf{u}(\mathbf{y}) - \mathbf{u}(\mathbf{x})}{|\mathbf{y} - \mathbf{x}|} \cdot \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|}$$

Peridynamics force:

$$-\nabla PD^\epsilon(\mathbf{u})(\mathbf{x}) = \frac{2}{|B_\epsilon(\mathbf{0})|} \int_{B_\epsilon(\mathbf{x})} \frac{J^\epsilon(|\mathbf{y} - \mathbf{x}|)}{\epsilon |\mathbf{y} - \mathbf{x}|} \partial_S f(|\mathbf{y} - \mathbf{x}| S^2) \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|} d\mathbf{y}$$

Linear peridynamics force: $f(r) = f'(0)r$ implies $\partial_S f(|\mathbf{y} - \mathbf{x}| S^2) = 2|\mathbf{y} - \mathbf{x}| S f'(0)$

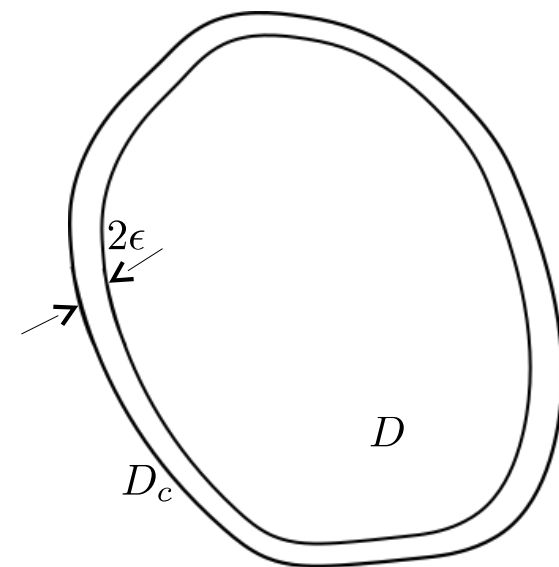
$$-\nabla PD_l^\epsilon(\mathbf{u})(\mathbf{x}) = \frac{2}{|B_\epsilon(\mathbf{0})|} \int_{B_\epsilon(\mathbf{x})} \frac{J^\epsilon(|\mathbf{y} - \mathbf{x}|)}{\epsilon |\mathbf{y} - \mathbf{x}|} (2S f'(0) |\mathbf{y} - \mathbf{x}|) \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|} d\mathbf{y}$$

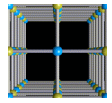
Elastic constants: The limiting equation, excluding cracking zone, is elastodynamics

$$\lambda = \mu = \frac{1}{4} f'(0) \int_0^1 r^2 J(r) dr, \quad \mathcal{G}_c = \frac{2}{\pi} f_\infty \int_0^1 r^2 J(r) dr$$

Influence function: Let $J : [0, 1] \rightarrow \mathbb{R}^+$ be bounded. For given ϵ

$$J^\epsilon(|\mathbf{x}|) = J(|\mathbf{x}|/\epsilon), \quad \mathbf{x} \in B_\epsilon(\mathbf{0})$$

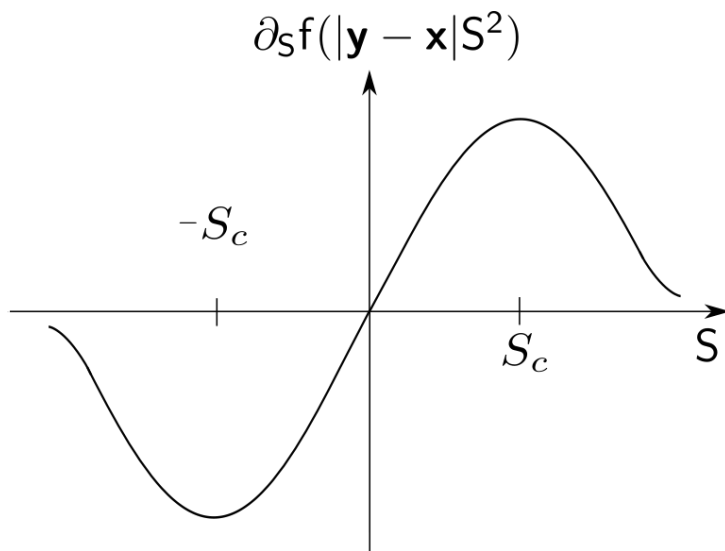
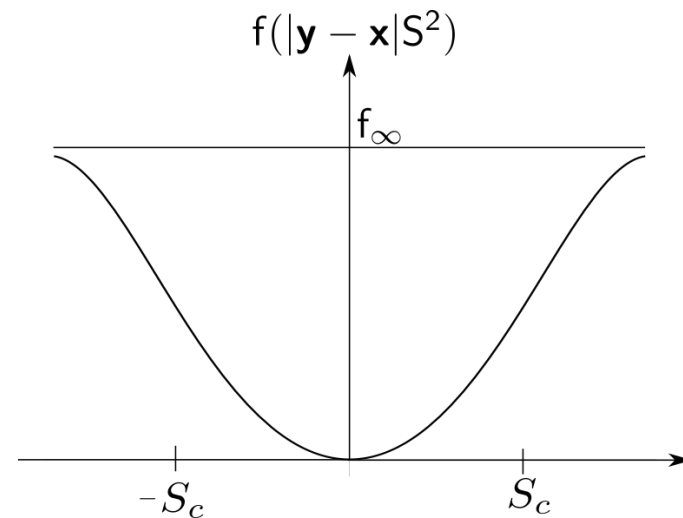
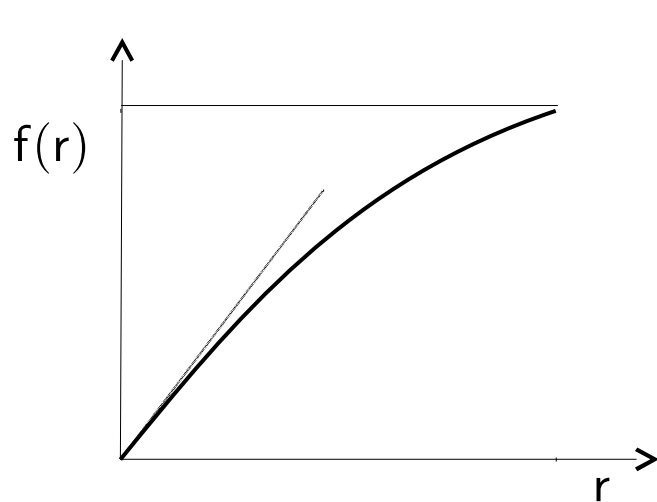




Potential and peridynamics force

2

f : smooth, bounded far away, and linear near origin (**Lipton 2014**¹)



- Critical strain: $S_c(\mathbf{y}, \mathbf{x}) = \frac{\bar{r}}{\sqrt{|\mathbf{y} - \mathbf{x}|}}$
- Weak form:

$$(\rho \ddot{\mathbf{u}}(t), \tilde{\mathbf{u}}) + \mathbf{a}^\epsilon(\mathbf{u}(t), \tilde{\mathbf{u}}) = (\mathbf{b}(t), \tilde{\mathbf{u}}) \quad \forall \tilde{\mathbf{u}} \in L^2$$
- If strain of sufficient bonds exceed S_c , coercivity of $\mathbf{a}^\epsilon$ is lost

Existence of solution

● In **Lipton 2014**¹ existence of solution in $C^2([0, T]; L^2(D, \mathbb{R}^d))$ is shown provided initial conditions are in $L^2(D, \mathbb{R}^d)$ and body force $\mathbf{b} \in C^1([0, T]; L^2(D, \mathbb{R}^d))$.

Proof follows once we show that peridynamic force is Lipschitz continuous in $L^2(D, \mathbb{R}^3)$, i.e. $\exists$ constant $L > 0$ independent of $\mathbf{u}, \mathbf{v} \in L^2(D, \mathbb{R}^3)$ such that

$$\| -\nabla PD^\epsilon(\mathbf{u}) - (-\nabla PD^\epsilon(\mathbf{v})) \|_{L^2} \leq \frac{L}{\epsilon^2} \|\mathbf{u} - \mathbf{v}\|_{L^2}$$

● Existence of solution in Hölder space is shown in **Jha and Lipton 2017**².

Let $\gamma \in (0, 1]$ and Hölder norm given by

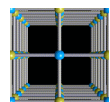
$$\|\mathbf{u}\|_{C^{0,\gamma}} := \sup_{\mathbf{x} \in D} |\mathbf{u}(\mathbf{x})| + \sup_{\mathbf{x}, \mathbf{y} \in D, \mathbf{x} \neq \mathbf{y}} \frac{|\mathbf{u}(\mathbf{x}) - \mathbf{u}(\mathbf{y})|}{|\mathbf{x} - \mathbf{y}|^\gamma}$$

$C^{0,\gamma}$ is space of functions which have bounded Hölder norm.

$$\| -\nabla PD^\epsilon(\mathbf{u}) - (-\nabla PD^\epsilon(\mathbf{v})) \|_{C^{0,\gamma}} \leq \frac{L_1 + L_2(\|\mathbf{u}\|_{C^{0,\gamma}} + \|\mathbf{v}\|_{C^{0,\gamma}})}{\epsilon^2} \|\mathbf{u} - \mathbf{v}\|_{C^{0,\gamma}}$$

[1] Robert Lipton (2014) Dynamic brittle fracture as a small horizon limit of peridynamics Journal of Elasticity 117 no. 1 2150

[2] Prashant K Jha and Robert Lipton (2017) Numerical analysis of nonlocal fracture models in Hölder space. arXiv preprint arXiv:1701.02818



Existence of solution

Outline of proof is as follows

1. Lipschitz continuity in Hölder space

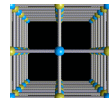
$$\| -\nabla PD^\epsilon(\mathbf{u}) - (-\nabla PD^\epsilon(\mathbf{v})) \|_{C^{0,\gamma}} \leq \frac{L_1 + L_2(\|\mathbf{u}\|_{C^{0,\gamma}} + \|\mathbf{v}\|_{C^{0,\gamma}})}{\epsilon^2} \|\mathbf{u} - \mathbf{v}\|_{C^{0,\gamma}}$$

2. Local existence of fixed point $y(t) = S(y(t))$ where

$$S(y(t)) = x_0 + \int_0^t F^\epsilon(y(\tau), \tau) d\tau, \quad x_0 = (\mathbf{u}_0, \mathbf{v}_0)$$

For given initial condition x_0 , we show that there exists $T' > 0$, independent of initial condition, such that unique fixed point exists for $t \in [-T', T']$.

3. Global existence of solution: For any given $T > 0$, using local existence theorem for small intervals we can show the existence of unique solution for all $t \in [-T, T]$.



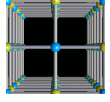
References

Theoretical analysis

Silling & Lehoucq 2008, Lipton, Silling & Lehoucq 20016
Aksoylu & Parks 2011, Aksoylu & Unlu 2014
Du & Gunzburger, Lehoucq & Zhou 2012, Du & Zhou 2011
Lipton 2014, Lipton 2015, Mengesha & Du 2013 Mengesha & Du 2014
Tian, Du & Gunzburger 2016, Emmrich & Puhst 2015
Emmrich & Puhst 2015 Dayal 2017

Numerical analysis and approximation

Silling & Askari 2005, Silling & Bobaru 2005, Bobaru et. al. 2009
Silling, Weckner, Askari & Bobaru 2010, Bobaru & Hu 2012
Chen & Gunzburger 2011, Du, Gunzburger, & Schiweitzer 2016
Guan & Gunzburger 2015, Tian, Du & Gunzburger 2016
Wildman & Gazonas 2014, Weckner & Emmrich 2005, Weckner 2009
Weckner & Abeyaratne 2005, Diehl, Lipton & Schiweitzer 2016



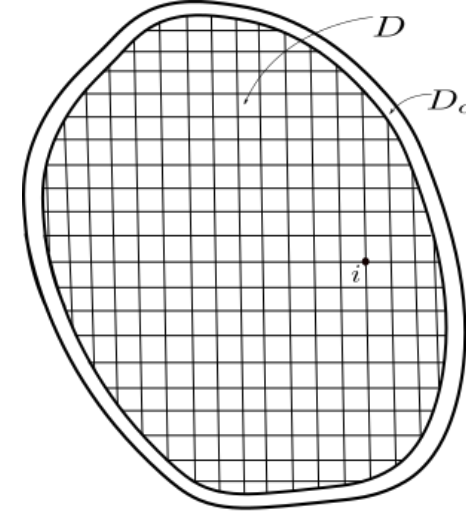
Finite difference approximation

$$D_h = D \cap h\mathbb{Z}^d, \mathbf{x}_i = \mathbf{i}h, t^k = k\Delta t$$

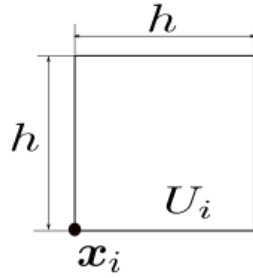
$(\hat{\mathbf{u}}_i^k, \hat{\mathbf{v}}_i^k)$ satisfies

$$\frac{\hat{\mathbf{u}}_i^{k+1} - \hat{\mathbf{u}}_i^k}{\Delta t} = \hat{\mathbf{v}}_i^k$$

$$\frac{\hat{\mathbf{v}}_i^{k+1} - \hat{\mathbf{v}}_i^k}{\Delta t} = -\nabla PD^\epsilon(\hat{\mathbf{u}}_h^k)(\mathbf{x}_i) + \mathbf{b}(\mathbf{x}_i, t^k)$$



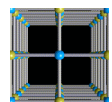
(a)



(b)

where $\hat{\mathbf{u}}_h^k, \hat{\mathbf{v}}_h^k$ are piecewise constant extension of discrete solution $\hat{\mathbf{u}}_i^k, \hat{\mathbf{v}}_i^k$

$$\hat{\mathbf{u}}_h^k(\mathbf{x}) := \sum_{\mathbf{i} \in \mathbb{Z}^d, \mathbf{i}h \in D} \hat{\mathbf{u}}_i^k \chi_{U_i}(\mathbf{x})$$



Convergence of finite difference approximation

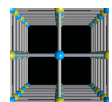
- Error E^k is given by

$$E^k = \|\hat{\mathbf{u}}_h^k - \mathbf{u}(t^k)\|_{L^2} + \|\hat{\mathbf{v}}_h^k - \mathbf{v}(t^k)\|_{L^2}$$

- If $\mathbf{u}, \mathbf{v} \in C^2([0, T]; C^{0,\gamma}(D, \mathbb{R}^d))$ then error satisfies following (**Jha & Lipton 2017¹**)

$$\sup_{k \leq T/\Delta t} E^k = O(\Delta t + h^\gamma/\epsilon^2)$$

- Similar rate of convergence is shown for general one step time discretization.



Convergence of finite difference approximation

Outline of proof:

1. Comparing exact solution $(\mathbf{u}(t^k), \mathbf{v}(t^k))$ with its piecewise constant projection $(\tilde{\mathbf{u}}_h^k, \tilde{\mathbf{v}}_h^k)$ given by

$$\tilde{\mathbf{u}}_h^k(\mathbf{x}) = \sum_{\mathbf{x}_i \in D \cap (h\mathbb{Z}^d)} \chi_{U_1}(\mathbf{x}) \tilde{\mathbf{u}}_i^k, \quad \tilde{\mathbf{u}}_i^k = \frac{1}{h^d} \int_{U_i} \mathbf{u}(\mathbf{y}, t^k) d\mathbf{y}$$

We can show $\|\mathbf{u}(t^k) - \tilde{\mathbf{u}}_h^k\|_{L^2} = O(h^\gamma)$.

2. Comparing $(\tilde{\mathbf{u}}_h^k, \tilde{\mathbf{v}}_h^k)$ with piecewise constant extension $(\hat{\mathbf{u}}_h^k, \hat{\mathbf{v}}_h^k)$ of discrete approximate solution.

- (1) Writing eridynamics equation for $(\tilde{\mathbf{u}}_h^k, \tilde{\mathbf{v}}_h^k)$ involving consistency error.

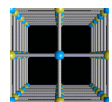
- (2) Estimating consistency error term. One of the error term is as follows

$$\| -\nabla P D^\epsilon(\mathbf{u}(t^k)) - (-\nabla P D^\epsilon(\tilde{\mathbf{u}}_h^k)) \|_{L^2} \leq \frac{L}{\epsilon^2} \|\mathbf{u}(t^k) - \tilde{\mathbf{u}}_h^k\|_{L^2} \leq \frac{L}{\epsilon^2} C h^\gamma$$

- (3) By substracting peridynamics equation corresponding to $(\tilde{\mathbf{u}}_h^k, \tilde{\mathbf{v}}_{kh})$

and $(\hat{\mathbf{u}}_h^k, \hat{\mathbf{v}}_h^k)$, we get bounds on $\|\hat{\mathbf{u}}_h^k - \tilde{\mathbf{u}}_h^k\|_{L^2} + \|\hat{\mathbf{v}}_h^k - \tilde{\mathbf{v}}_h^k\|_{L^2}$

$$3. E^k \leq \|\mathbf{u}(t^k) - \tilde{\mathbf{u}}_h^k\|_{L^2} + \|\mathbf{v}(t^k) - \tilde{\mathbf{v}}_h^k\|_{L^2} + \|\tilde{\mathbf{u}}_h^k - \hat{\mathbf{u}}_h^k\|_{L^2} + \|\tilde{\mathbf{v}}_h^k - \hat{\mathbf{v}}_h^k\|_{L^2}$$



One dimensional model

Objective:

- To see if we can improve the rate of convergence h^γ/ϵ^2 where $\gamma \in (0, 1]$.
- Rate of convergence of nonlinear peridynamics to elastodynamics.

We show in **Jha & Lipton 2017**¹ that if $u \in C^2([0, T]; C^4(D, \mathbb{R}))$ then solution of nonlinear peridynamic converges to the elastodynamic solution, i.e.

$$u^\epsilon \rightarrow u \quad \text{in } H_0^1(D)$$

at the rate ϵ uniformly in time $t \in [0, T]$.

We follow the quadrature based FE approximation in **Tian & Du 2014**² and **Tian, Du & Gunzburger 2016**³. Let $\mathcal{I}_h[u]$ is given by

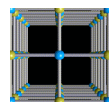
$$\mathcal{I}_h[u](x) = \sum_{i, x_i \in D} u(x_i) \phi_i(x)$$

[1] Prashant K. Jha and Robert Lipton (2017) Numerical convergence of nonlinear nonlocal continuum models to local elastodynamics. arXiv:1707.00398

[2] X. Tian and Qiang Du (2014) Asymptotically compatible schemes and applications to robust discretization of nonlocal models.

SIAM Journal on Numerical Analysis, 52(4):1641–1665.

[3] X. Tian, Qiang Du and Max Gunzburger (2016) Asymptotically compatible schemes for the approximation of fractional laplacian and related onlocal diffusion problems on bounded domains. Advances in Computational Mathematics, 42(6):1363–1380.



One dimensional model

Let $\hat{u}_i^{\epsilon,k}$ denote the approximate displacement at $x_i = ih \in D, i \in \mathbb{Z}$ and $t^k = k\Delta t$. It satisfies

$$\frac{\hat{u}_i^{\epsilon,k+1} - 2\hat{u}_i^{\epsilon,k} + \hat{u}_i^{\epsilon,k-1}}{\Delta t^2} = -\nabla P D^\epsilon(\hat{u}_h^{\epsilon,k})(x_i) + b(t^k, x_i)$$

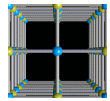
with initial condition as $\hat{u}_i^{\epsilon,0} = u_0(x_i)$. For initial condition on velocity, we use

$$\frac{\hat{u}_i^{\epsilon,1} - \hat{u}_i^{\epsilon,-1}}{2\Delta t} = v_0(x_i)$$

and use above to start the time iteration for $k = 0$.

$\hat{u}_h^{\epsilon,k}$ is the extension of discrete set $\hat{u}_i^{\epsilon,k}$ using linear interpolation functions ϕ_i . It is given by

$$\hat{u}_h^{\epsilon,k}(x) = \sum_{i \in \mathbb{Z}, ih \in [0,1]} \hat{u}_i^{\epsilon,k} \phi_i(x)$$



Consistency error and stability

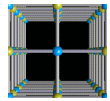
We approximate the peridynamic force $-\nabla \text{PD}^\epsilon(u)$ by $-\nabla \text{PD}^\epsilon(\mathcal{I}_h[u])$. Consistency error satisfies

$$\sup_{i, x_i \in D} |-\nabla \text{PD}^\epsilon(\mathcal{I}_h[u])(x_i) - (-\nabla \text{PD}^\epsilon(u)(x_i))| = O(h/\epsilon) \quad \leftarrow \text{comparing this with } O(h^\gamma/\epsilon^2)$$

$$\sup_{i, x_i \in D} |-\nabla \text{PD}^\epsilon(\mathcal{I}_h[u])(x_i) - \mathbb{C}u_{xx}(x_i)| = O(\epsilon) + O(h/\epsilon) \quad \leftarrow \text{rate of convergence to elastodynamics depends on } h/\epsilon.$$

Stability: For linear peridynamics, we obtain following stability condition, for Central difference scheme

$$\Delta t \leq \frac{h}{\sqrt{\mathbb{C} + 2f'(0)\frac{Mh^2}{\epsilon^2}}} \quad \text{where} \quad \mathbb{C} = \frac{2f'(0)}{\epsilon^2} \int_0^1 J(|z|)zdz$$



Numerical verification: h-convergence

- For fixed ϵ we expect the rate of convergence to be $O(h)$.

$$u_0(x) = a \exp[-(0.5 - x)^2 / \beta], v_0(x) = 0$$

$$a = 0.005, \beta = 10^{-5}, \text{ time domain} = [0, 1.7]$$

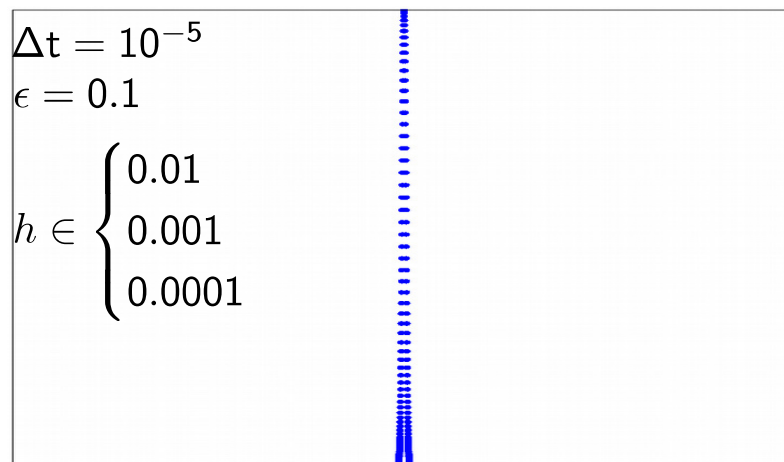
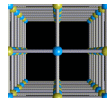


Table: LPD refers to linear peridynamics and NPD refer to nonlinear peridynamics. Superscript "1" corresponds to L^2 norm and "2" corresponds to sup norm.

Time step	LPD ¹	NPD ¹	LPD ²	NPD ²
6000	1.6416	1.6419	1.4204	1.4204
51500	1.3098	1.3106	1.3312	1.3331
104000	1.1504	1.1482	1.5155	1.5557
147000	1.1364	1.1262	1.6027	1.5215
165000	1.2611	1.2632	1.5496	1.6055

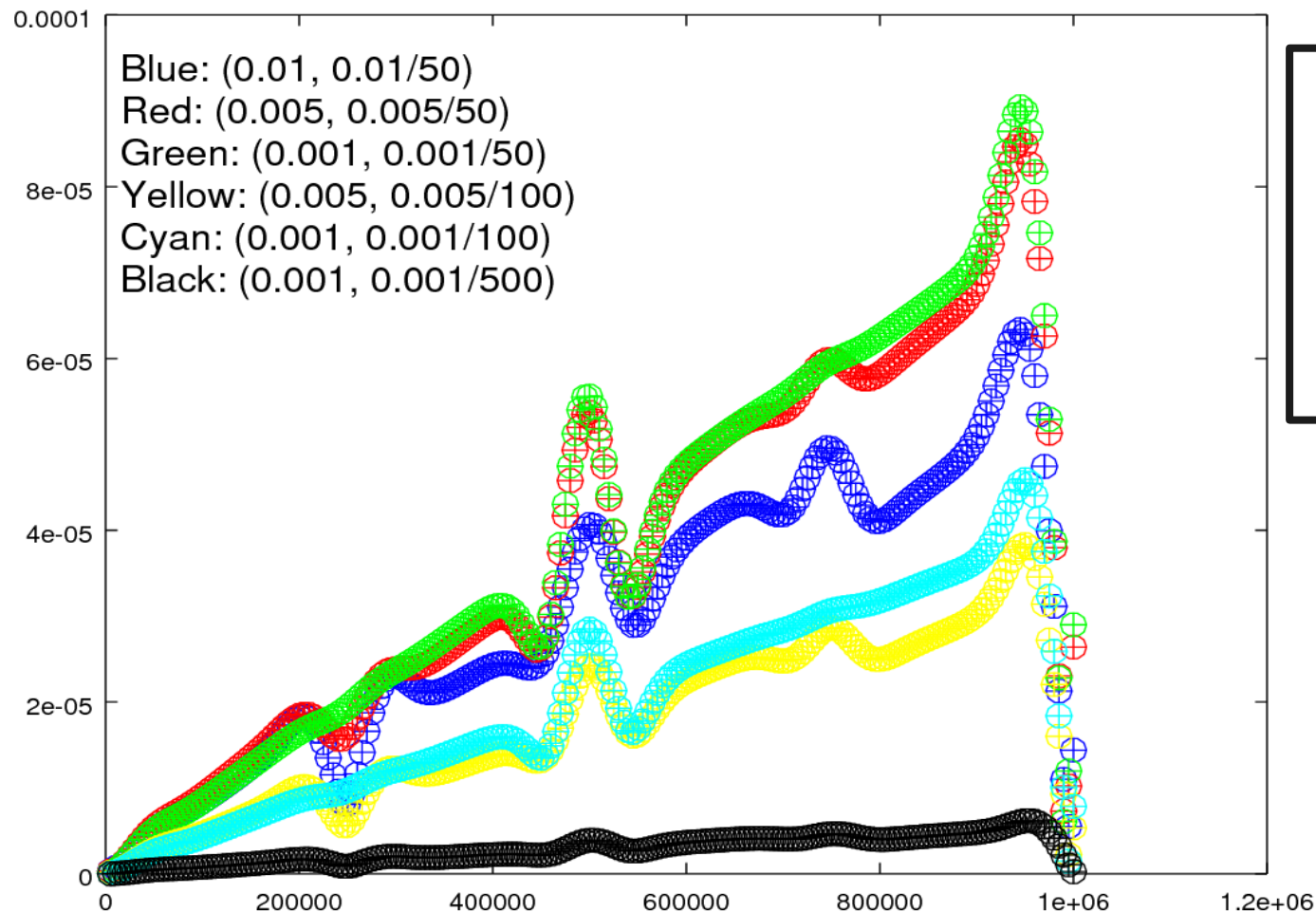
- Similar results have been observed for multiple simulations.



Numerical verification: m-convergence

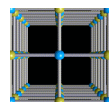
$$u_0(x) = a \exp[-(0.25 - x)^2/\beta - (0.75 - x)^2/\beta], v_0(x) = 0, a = 0.001, \beta = 0.003.$$

Let $\Delta t = 10^{-6}$, time domain is $[0, 1]$.



h has to decrease at faster rate for decreasing ϵ to reduce the error between peridynamic and elastodynamic solution.

Plot of time step k vs $\|u_{\text{peri}} - u_{\text{elasto}}\|_{L^2}$. “+” corresponds to LPD and “o” corresponds to NPD.



Finite element approximation using linear interpolation¹⁴

Using weak formulation and applying finite element approximation, we can further improve the rate of convergence of error. Consider following

$$\begin{aligned}(\ddot{\mathbf{u}}(t), \tilde{\mathbf{u}}) + a^\epsilon(\mathbf{u}(t), \tilde{\mathbf{u}}) &= (\mathbf{b}(t), \tilde{\mathbf{u}}) & \tilde{\mathbf{u}} &\in L_0^2(D, \mathbb{R}^d) \\(\ddot{\mathbf{u}}_l(t), \tilde{\mathbf{u}}) + a_l^\epsilon(\mathbf{u}_l(t), \tilde{\mathbf{u}}) &= (\mathbf{b}(t), \tilde{\mathbf{u}}) & \tilde{\mathbf{u}} &\in L_0^2(D, \mathbb{R}^d)\end{aligned}$$

where

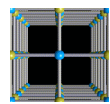
$$a^\epsilon(\mathbf{u}, \mathbf{v}) = \frac{2}{\epsilon^{d+1}\omega_d} \int_D \int_D J^\epsilon(|\mathbf{y} - \mathbf{x}|) f'(|\mathbf{y} - \mathbf{x}| S(\mathbf{u})^2) |\mathbf{y} - \mathbf{x}| S(\mathbf{u}) S(\mathbf{v}) d\mathbf{x} d\mathbf{y}$$

$$a_l^\epsilon(\mathbf{u}, \mathbf{v}) = \frac{2}{\epsilon^{d+1}\omega_d} \int_D \int_D J^\epsilon(|\mathbf{y} - \mathbf{x}|) f'(0) |\mathbf{y} - \mathbf{x}| S(\mathbf{u}) S(\mathbf{v}) d\mathbf{x} d\mathbf{y}$$

$$S(\mathbf{u}) = \frac{\mathbf{u}(\mathbf{y}) - \mathbf{u}(\mathbf{x})}{|\mathbf{y} - \mathbf{x}|} \cdot \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|}$$

Energy is defined as

$$e^\epsilon(\mathbf{u}(t)) = \frac{1}{2} \rho \|\dot{\mathbf{u}}(t)\|_{L^2}^2 + a^\epsilon(\mathbf{u}(t), \mathbf{u}(t))$$



Finite element approximation using linear interpolation 15

● a_l^ϵ is coercive and bounded. If displacement field is such that $S(\mathbf{u}) \leq C < S_c$ for all $|\mathbf{y} - \mathbf{x}| \leq \epsilon$, we can show the coercivity of a^ϵ .

● For linear peridynamics, following **Baker 1976**¹, **Grote & Schötzau 2009**², **Karaa 2012**³, **Guan & Gunzburger 2015**⁴, we can obtain the condition on Δt .

● We can show the stability of semi-discrete scheme for linear Peridynamics and under the assumption of $S(\mathbf{u}) < S_c$ for nonlinear peridynamics.

● Using bound on energy, we can show for any $t_1 > t_2, t_1, t_2 \in [0, T]$, we have

$$\|u_h(t_1) - u_h(t_2)\|_{L^2} \leq |t_1 - t_2| \left[\sup_{t \in [0, T]} e^\epsilon(u_h(t)) \right]$$

● If exact solution $(\mathbf{u}(t), \mathbf{v}(t)) \in H_0^2(D, \mathbb{R}^d)$ then error $E^k = \|\hat{\mathbf{u}}^k - \mathbf{u}(t^k)\|_{L^2}$ satisfies

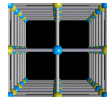
$$\sup_k E^k = O(\Delta t + h^2/\epsilon^2)$$

[1] G. A. Baker (1976) Error estimates for finite element methods for second order hyperbolic equations. SIAM journal on numerical analysis, 13(4):564–576.

[2] M. J. Grote and D. Schötzau (2009) Optimal error estimates for the fully discrete interior penalty dg method for the wave equation. Journal of Scientific Computing, 40(1):257–272.

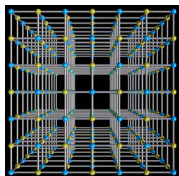
[3] S. Karaa (2012) Stability and convergence of fully discrete finite element schemes for the acoustic wave equation. Journal of Applied Mathematics and Computing, 40(1-2):659–682.

[4] Q. Guan and M. Gunzburger (2015) Stability and accuracy of time stepping schemes and dispersion relations for a nonlocal wave equation. Numerical Methods for Partial Differential Equations, 31(2):500–516.



Future works

- Numerical analysis of state-based nonlinear model
- Nonlocal in time, i.e. damage models
- Numerical implementation of state-based and damager models



Thank you!